Predicting Asteroid Orbital Proximity Using Machine Learning

A comprehensive exploratory analysis of near-Earth asteroid orbital behavior using statistical learning and explainable AI techniques.

Background and Motivation

Why Study Near-Earth Asteroids?

Near-Earth asteroids (NEAs) possess orbital trajectories that pass relatively close to Earth's orbit, making them objects of critical scientific importance. Monitoring these celestial bodies is essential for understanding orbital behavior, planetary defense strategies, and potential impact risk assessment.

The increasing availability of astronomical observation data has created unprecedented opportunities for applying statistical analysis and machine learning techniques to space science problems.

Key Challenge

Asteroid orbital systems are highly complex, with relationships that are nonlinear, highly correlated, and statistically irregular. Traditional linear methods struggle to capture these intricate patterns.

Problem Statement and Objectives

Primary Aim

Predict transformed asteroid orbital proximity (MOID_log) using asteroid orbital and physical characteristics

Secondary Goals

- Study relationships among orbital variables
- Investigate nonlinear orbital behavior
- Compare linear and nonlinear models
- Identify influential features

Interpretation

Interpret model behavior using explainable AI techniques (SHAP analysis)

What is MOID?

Minimum Orbit Intersection Distance (MOID) represents the minimum possible distance between the orbital path of Earth and that of an asteroid. Smaller MOID values indicate closer orbital approaches and greater relevance in proximity analysis and hazard assessment studies.

Dataset Description and Preprocessing

Dataset Characteristics

- 4,687 asteroid observations
- 14 predictor variables
- Target: MOID_log (transformed)
- Source: Near-Earth asteroid catalog

Key Variables

- Orbital geometry: Eccentricity, Inclination, Semi-Major Axis
- Orbital motion: Period, Mean Motion
- Physical characteristics: Diameter, Magnitude
- Proximity measures: Perihelion, Aphelion

Preprocessing Steps

1. Removed non-predictive identifier columns
2. Removed incomplete observations
3. SelRetained numerical orbital variables
4. Created modeling dataset

Quality Control

Range validation confirmed physically meaningful values: Eccentricity (0.007-0.96), Inclination (0.01-75°), MOID (0.000002-0.48 AU)

Feature Engineering

Log Transformation

Applied log(1 + MOID) to reduce skewness from 1.47 to 1.30

Threat Score

Engineered custom hazard metric combining size, velocity, and proximity

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MOID Transformation

Initial inspection revealed strong positive skewness in raw MOID distribution. Most asteroids had small MOID values, while fewer extended to larger distances.

Transformation applied:

$$\text{MOID\_log} = \log(1 + \text{MOID})$$

The addition of 1 ensured safe transformation of values near zero. This reduced skewness, compressed extreme values, and stabilized the target distribution.

Threat Score Engineering

Created physically meaningful hazard potential metric:

$$\text{Threat Score} = \frac{\text{Diameter} \times \text{Velocity}}{\text{Miss Distance}}$$

Rationale: Dangerous asteroids depend on proximity, size, and speed. Large, fast-moving asteroids passing close to Earth are more hazardous than small, slow ones at similar distances.

Exploratory Data Analysis (EDA)

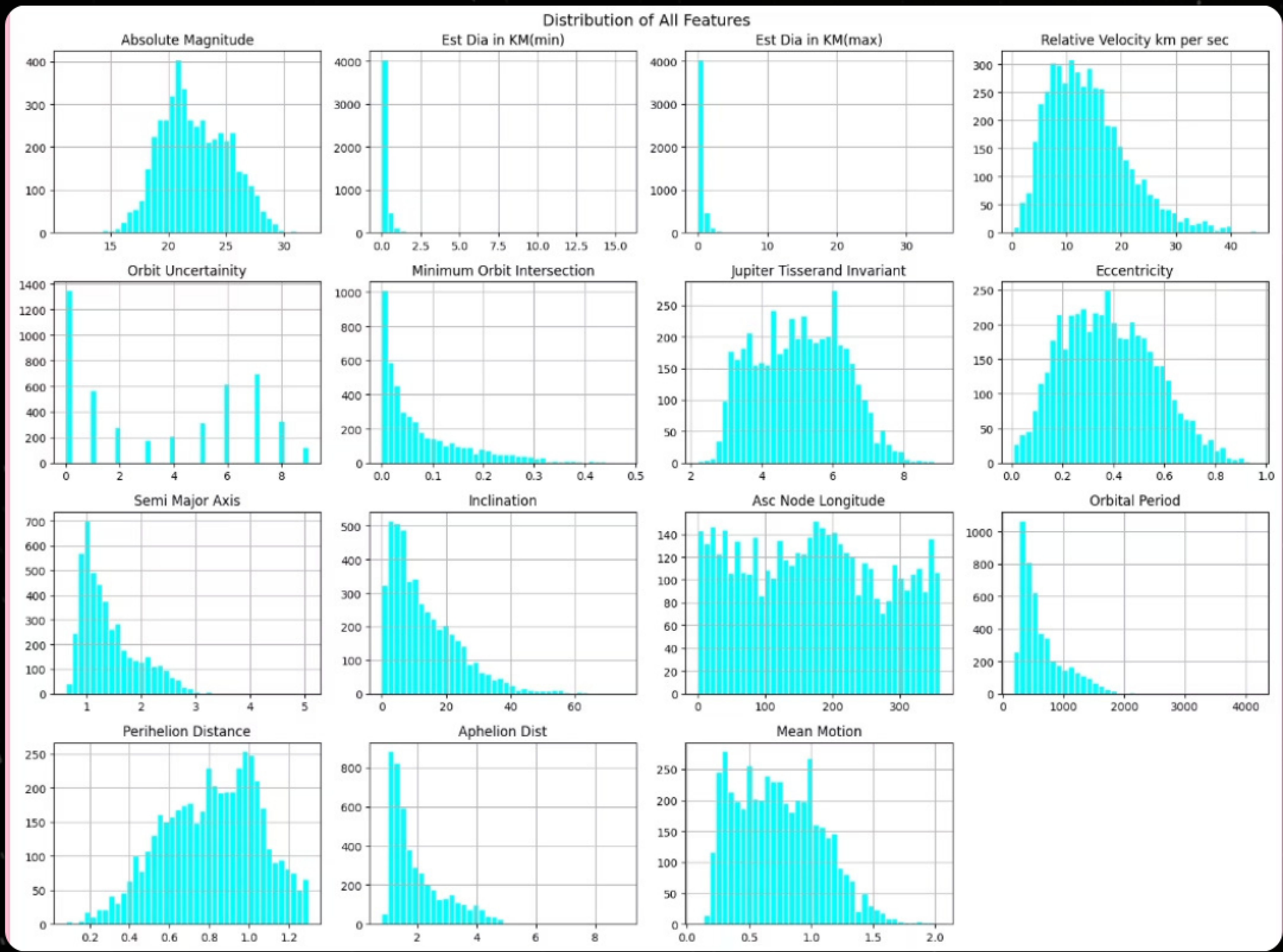
Purpose of EDA

- Understand the structure of asteroid orbital data
- Study distributions and skewness of variables
- Identify relationships among orbital parameters
- Detect nonlinear relationships and statistical irregularities

Key Questions Investigated

- Which orbital variables most strongly influence MOID_log?
- Are orbital relationships linear or nonlinear?
- Is multicollinearity present among predictors?
- Do the variables show skewness or heteroscedasticity?

Understanding the statistical structure and spread of asteroid features

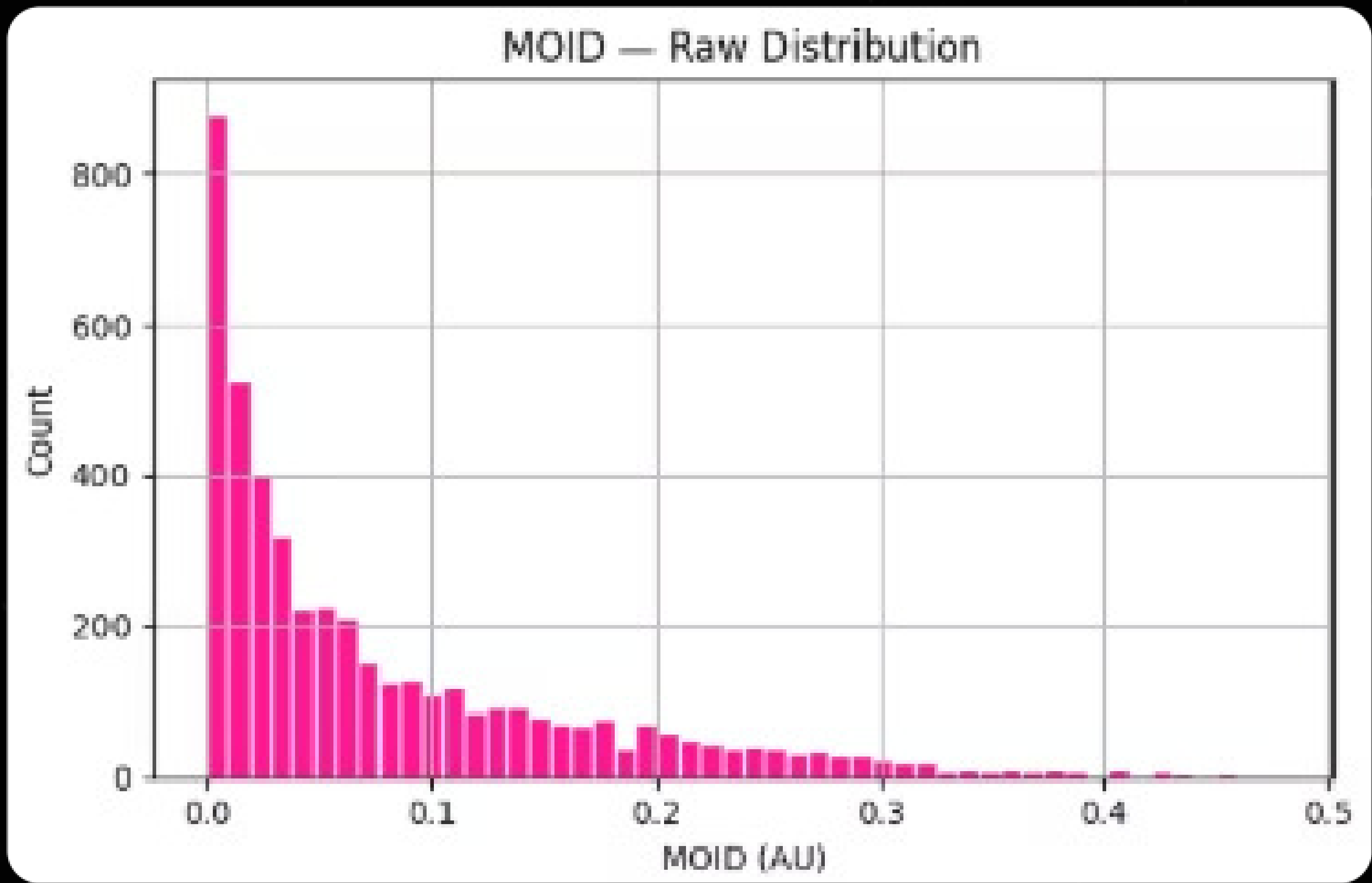


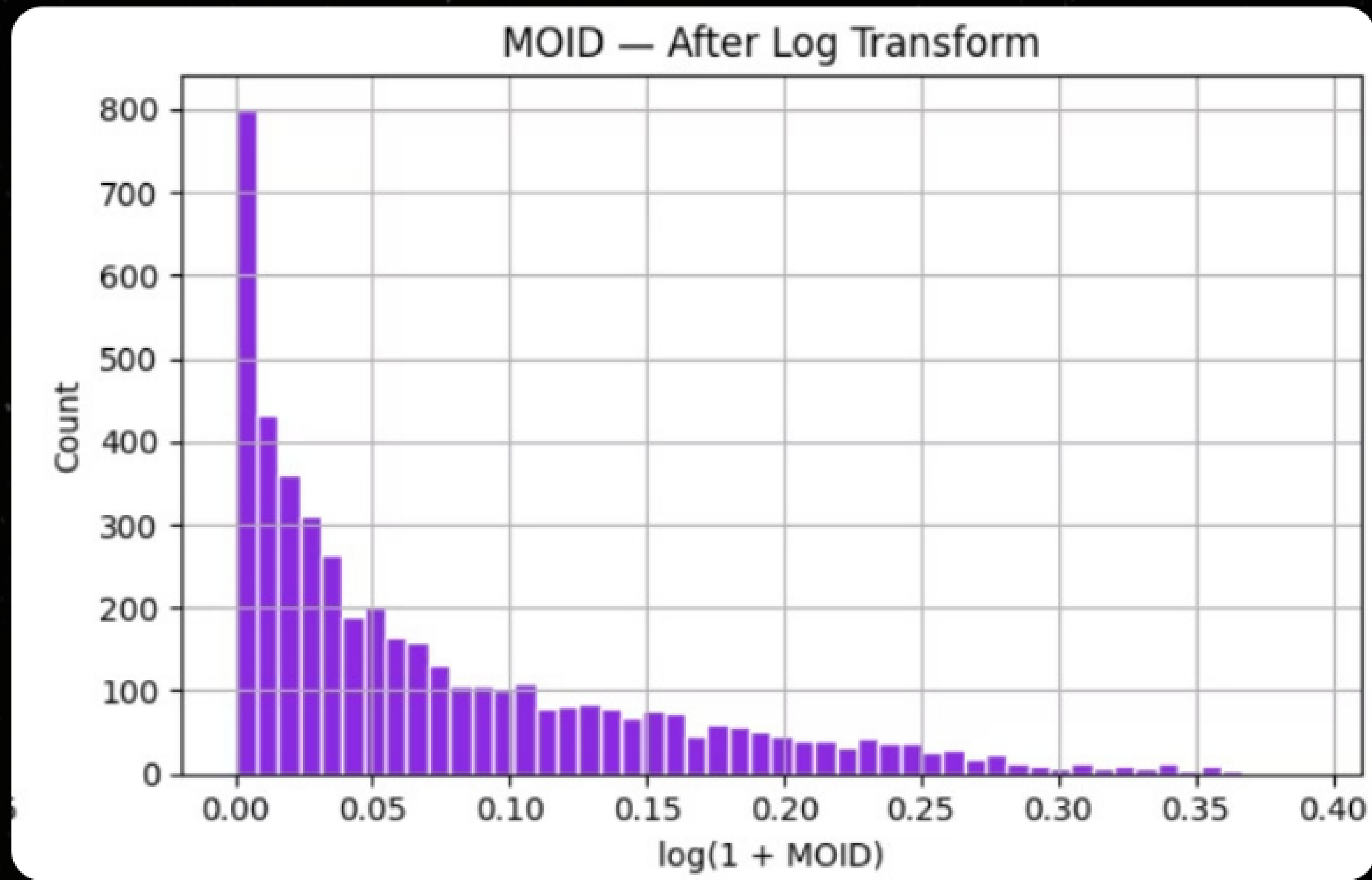
Key Observations

- Most orbital variables showed positive skewness
- Several variables contained long right tails and extreme values
- Diameter and orbital period distributions were heavily concentrated near smaller values
- Inclination indicated most asteroids orbit near Earth’s orbital plane
- Orbit Uncertainty displayed discrete spike-like behavior

The dataset exhibited strong skewness and statistical irregularities, suggesting that nonlinear modelling approaches may be more suitable than simple linear methods.

MOID Transformation Analysis





Why Transform**ation Was Needed**

- Raw MOID distribution showed strong positive skewness
- Most asteroids had very small MOID values, while fewer extended toward large distances
- Highly skewed targets can negatively affect regression performance and variance stability

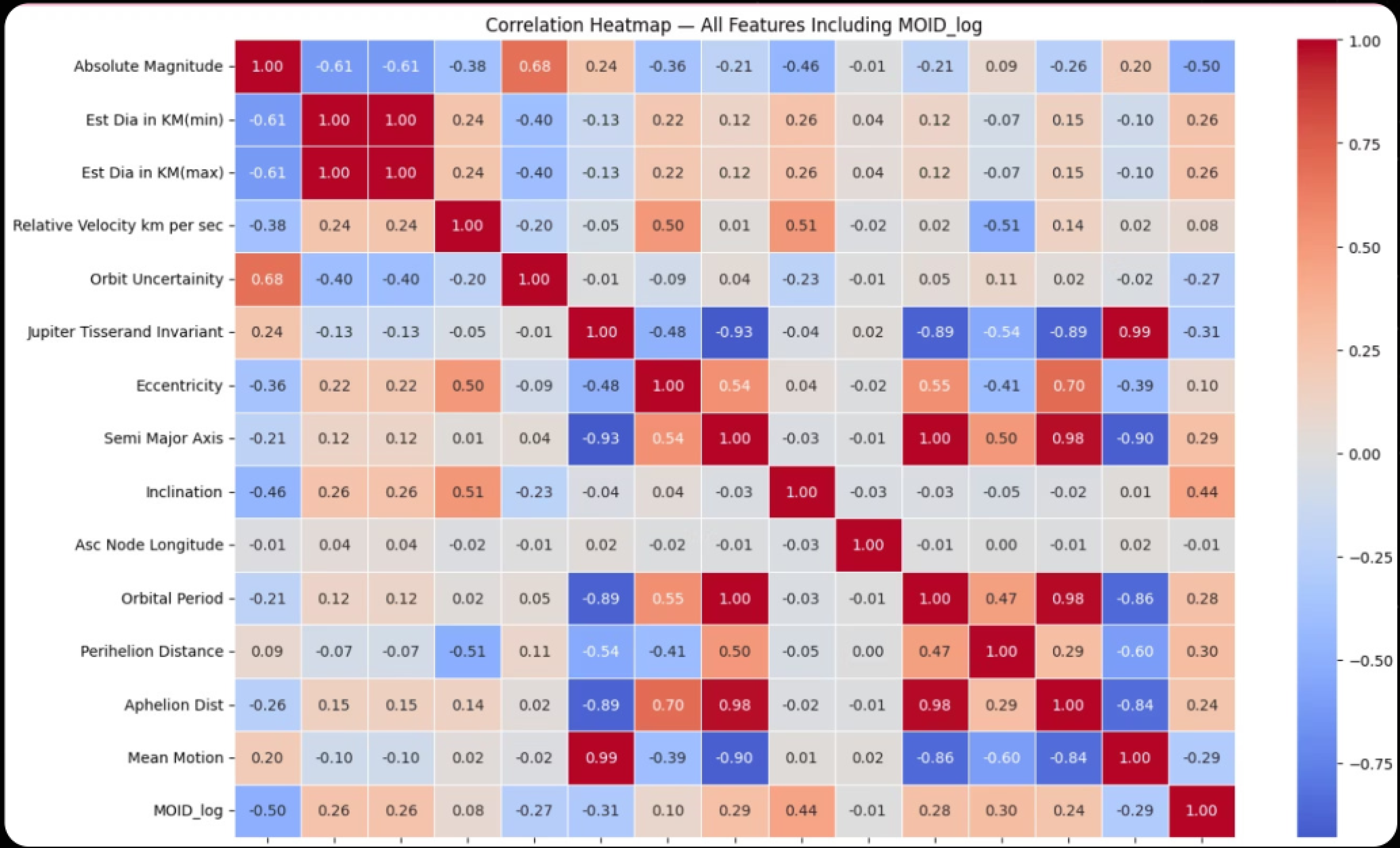
Transformation Applied

Make this visually highlighted:

$MOID_{l}og = log(1 + MOID)MOID\_log = \log(1 + MOID)$
MOID_log=log(1+MOID)

Results/insight

The original MOID distribution was heavily right-skewed, with most observations concentrated near smaller values. A logarithmic transformation was applied to reduce skewness, compress extreme values, and improve modelling stability before regression analysis.



Correlation Analysis of Orbital Variables

Major Observations

- Strong relationships were observed among several orbital variables
- Absolute Magnitude and Inclination showed notable relationships with MOID_log
- Semi-Major Axis, Orbital Period, and Aphelion Distance exhibited very high correlations
- Diameter min and max were almost perfectly correlated

Model Preparation and Data Splitting

Preparing the dataset for regression and ensemble learning models

X=Predictor Variables, Y=MOID_log

Train-Test Split

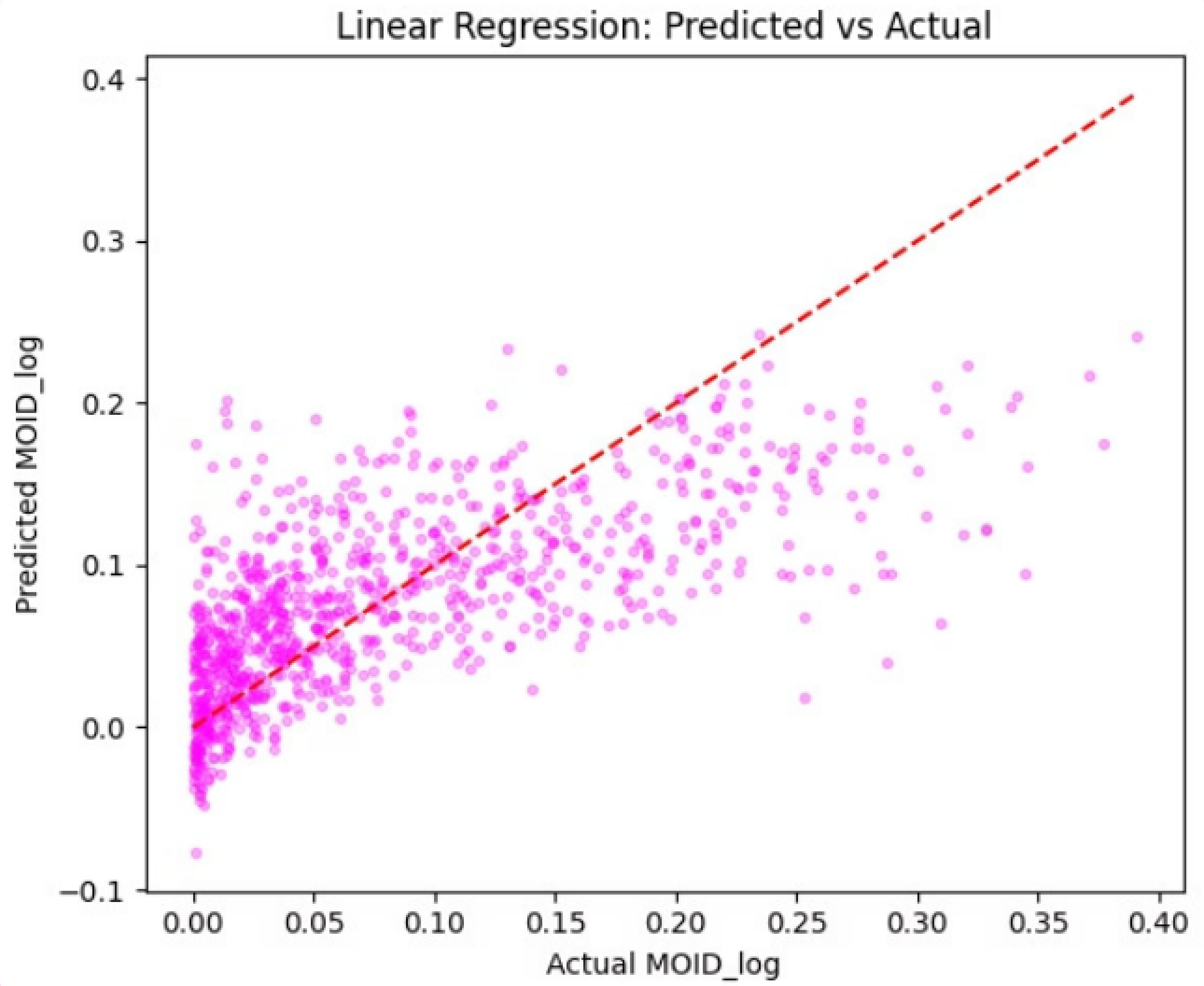
- 80% training data
- 20% testing data
- Random state fixed for reproducibility

Feature Scaling

- StandardScaler applied to numerical features
- Scaling performed using training data only
- Prevented data leakage during testing

Feature scaling ensured that variables with large numerical ranges did not dominate the learning process in regression-based models.

Final Dataset: 4,687 observations | 14 predictor variables | Target: MOID_log



Linear Regression — Baseline Model

Metric	Value
R^2	0.4542
RMSE	0.0600
MAE	0.0443

Major Findings

- Linear Regression captured moderate relationships within the dataset
- The model explained approximately 45% of MOID_log variance
- Predictions showed substantial spread away from the ideal diagonal line
- The model struggled to capture nonlinear orbital behaviour

“Linear regression was useful, but insufficient.”

Ridge and Lasso Regression

Lasso Regression

Metric	Value
R^2	0.4401
RMSE	0.0607
MAE	0.045

Ridge Regression

Metric	Value
R^2	0.4538
RMSE	0.0600
MAE	0.0443

Key Findings

- Lasso Regression performed automatic feature selection
- Several orbital variables received zero coefficients and were effectively removed
- Model performance decreased slightly compared to Ridge and Linear Regression

Key Findings

- Ridge Regression applied L2 regularization to stabilize coefficients
- Performance remained nearly identical to Linear Regression
- Regularization reduced coefficient instability caused by multicollinearity

Regularization improved coefficient stability and feature selection, but linear models still struggled to capture the nonlinear orbital relationships present in the dataset.

Random Forest Regression Metrics Box

Metric	Value
R^2	0.7640
RMSE	0.0394
MAE	0.0247

1

Major Findings

- Random Forest substantially outperformed all linear regression models
- The model captured nonlinear orbital relationships effectively
- Predictions aligned much more closely with actual MOID_log values
- Ensemble learning significantly improved predictive accuracy

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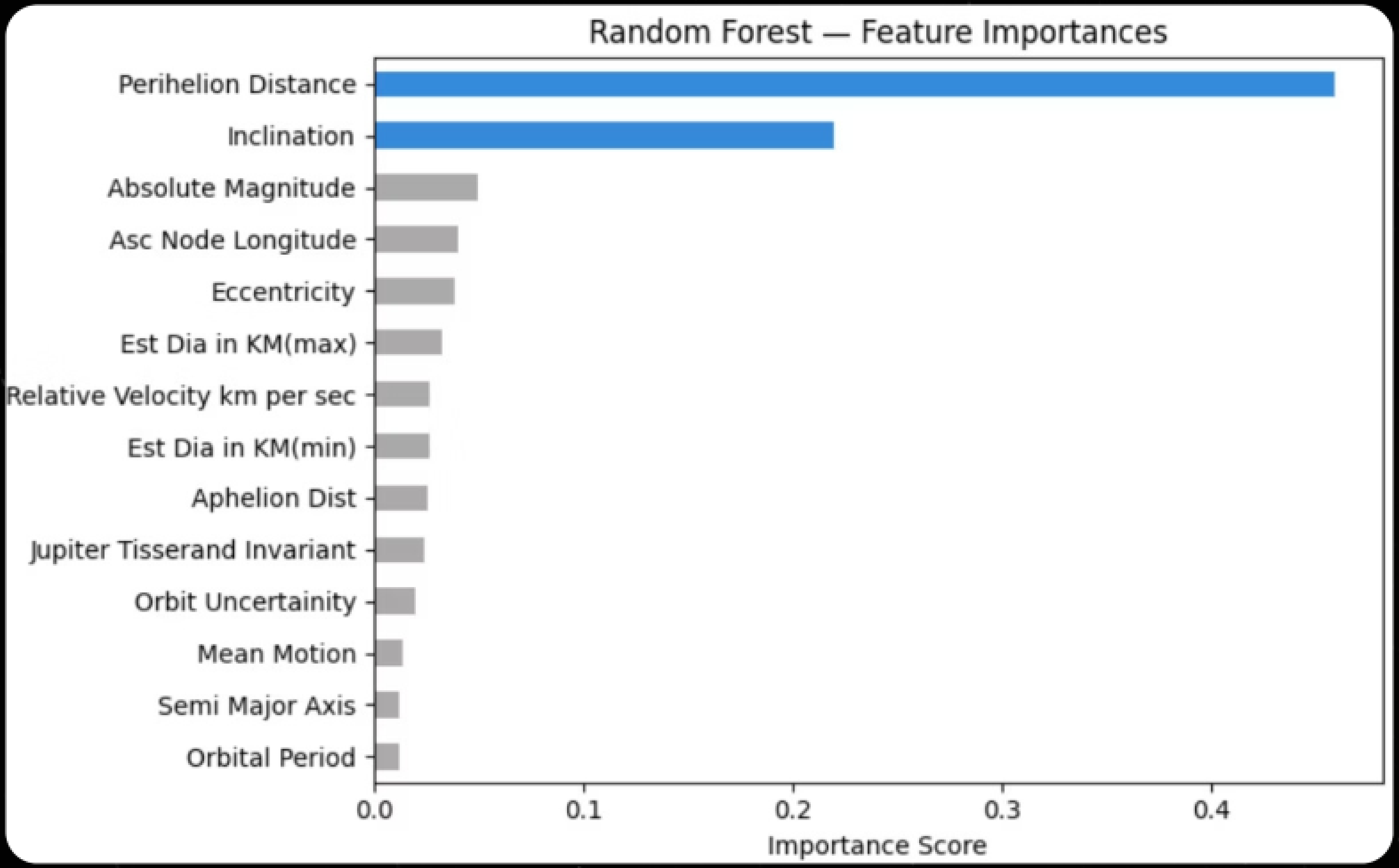
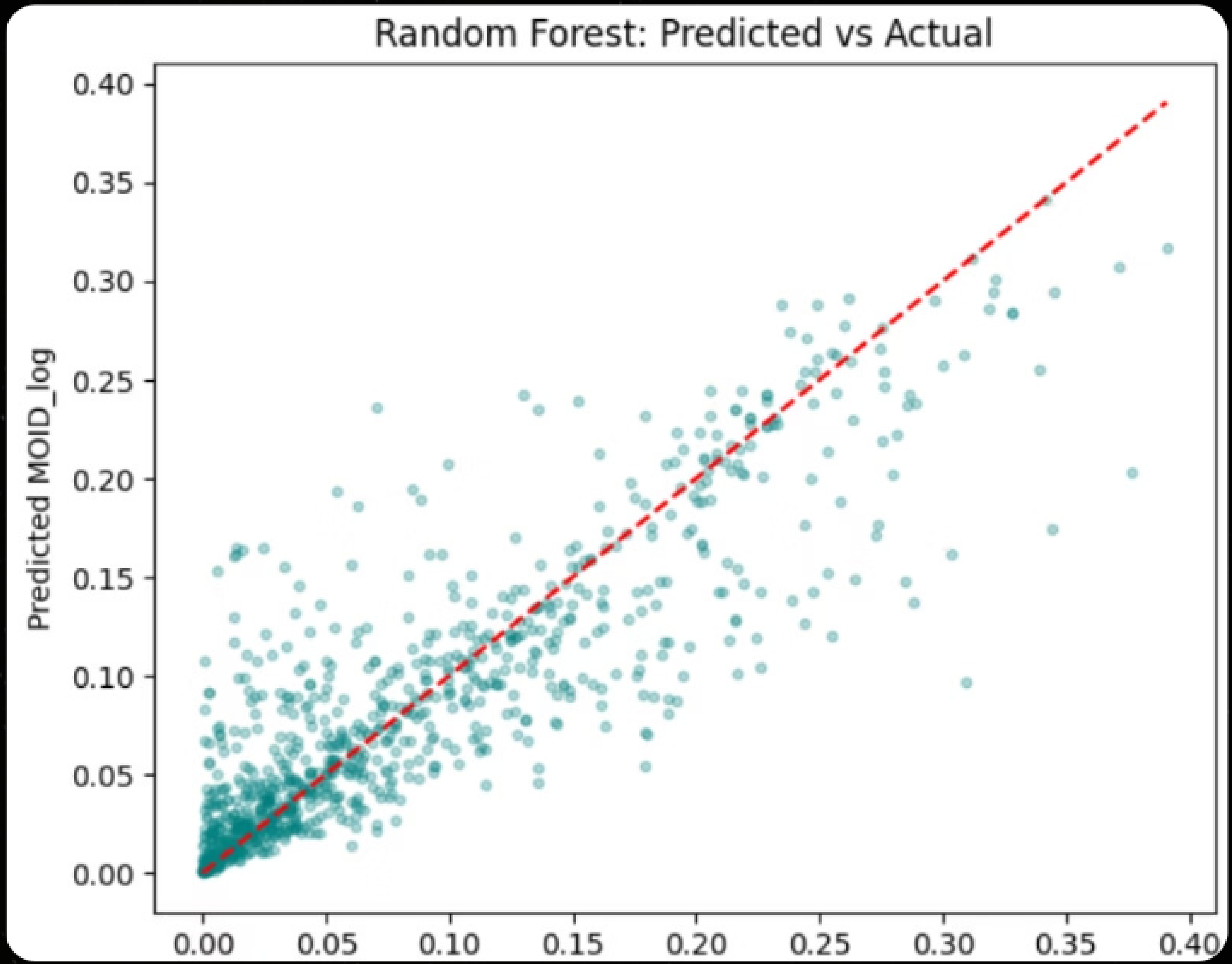
Most Influential Variables

- Perihelion Distance
- Inclination
- Absolute Magnitude

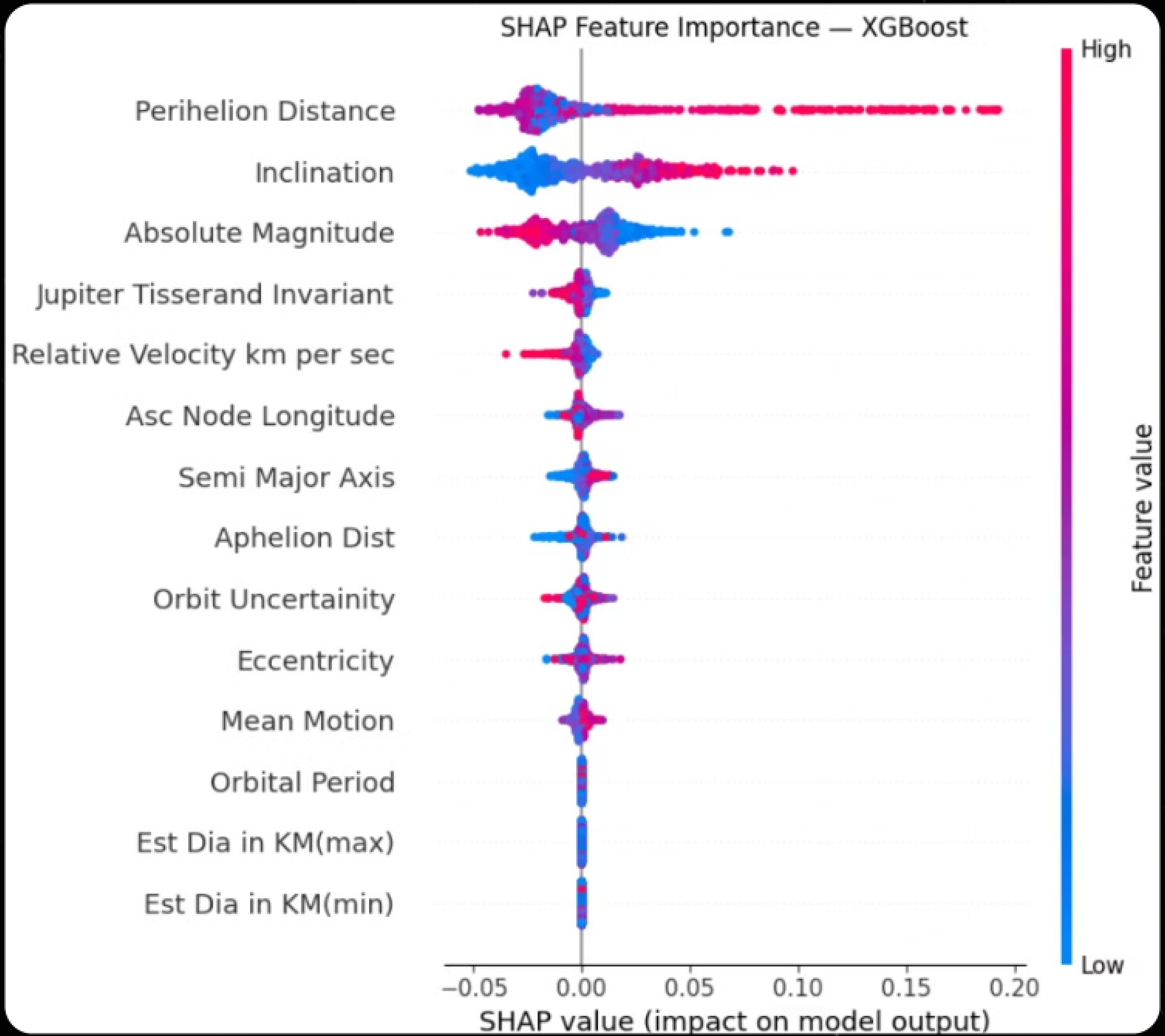


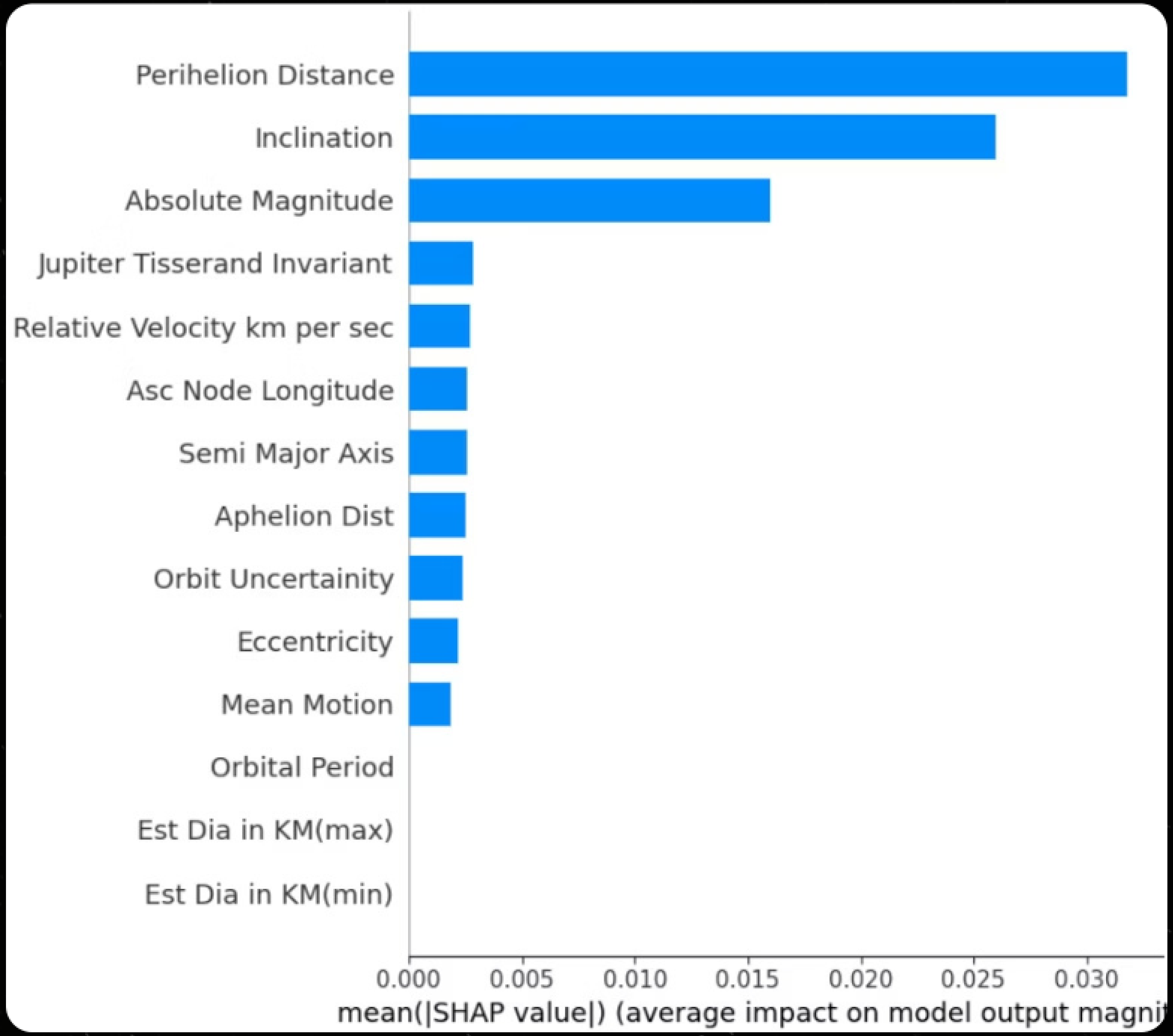


Important Insight

The strong improvement in performance confirmed that asteroid orbital proximity relationships are highly nonlinear and better captured through ensemble learning methods.

XGBoost Regression





Interpreting nonlinear orbital behaviour using explainable ensemble learning

Metrics Box

Metric	Value
R^2	0.7390
RMSE	0.0415
MAE	0.0277

Major Findings

- XGBoost achieved strong predictive performance close to Random Forest
- The model successfully captured nonlinear orbital relationships
- SHAP analysis enabled interpretation of feature contributions at both global and local levels

Feature Importance and SHAP Analysis

Perihelion Distance

Most Important: Represents closest point of asteroid's orbit to Sun. Dominance suggests orbital geometry near inner Solar System strongly influences Earth proximity

Inclination

Second Most Important: Orbital tilt relative to Earth's orbital plane. Highly inclined orbits less likely to closely align with Earth's path

Other Contributors

Absolute Magnitude, Eccentricity, Relative Velocity, Jupiter Tisserand Invariant showed moderate importance

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SHAP Explainability Insights

Directional Influence

- High perihelion distance → larger MOID_log (farther separation)
- High inclination → larger MOID_log (less aligned orbits)
- Lower magnitude (brighter/larger) → different prediction shifts

Scientific Validation

SHAP analysis confirmed EDA findings: orbital geometry variables (Perihelion Distance, Inclination) dominated all other features as primary determinants of Earth orbital proximity

Modeling Results: Linear vs Nonlinear Methods

0.454

Linear Regression R^2

Explained 45.4% of variance

0.764

Random Forest R^2

Explained 76.4% of variance

0.739

XGBoost R^2

Explained 73.9% of variance

Performance Comparison Summary

Model	R^2	RMSE	MAE
Linear	0.454	0.0600	0.0443
Ridge	0.454	0.0600	0.0443
Lasso	0.440	0.0607	0.0450
Random Forest	**0.764**	**0.0394**	**0.0247**
XGBoost	0.739	0.0415	0.0277

Key Finding: Tree-based nonlinear ensemble models substantially outperformed linear regression, confirming EDA observations of nonlinear orbital relationships

Conclusions and Future Directions

01

Nonlinear Relationships Confirmed

Scatterplot analysis revealed curved, funnel-shaped patterns that correlation alone could not detect

02

Ensemble Methods Superior

Random Forest explained 76.4% of variance vs 45.4% for linear regression

03

Orbital Geometry Dominant

Perihelion Distance and Inclination identified as most influential predictors

04

Explainable AI Validated

SHAP analysis provided meaningful scientific interpretation of orbital behavior

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Future Research Directions

Additional Features

Incorporate gravitational perturbations from other planets and solar radiation pressure effects

Time Series Analysis

Model temporal evolution of orbital parameters for long-term proximity forecasting

Impact Risk Integration

Combine proximity predictions with impact probability models for comprehensive hazard assessment

> *"Machine learning methods, particularly ensemble tree-based models, provide powerful tools for analyzing and predicting complex asteroid orbital behavior"*